Advanced Trigonometry

Section 1: Reciprocal & Pythagorean Identities

1 Solve the equation $2\tan^2\theta + \sec\theta - 1 = 0$ for $0^\circ \le \theta \le 360^\circ$.

a. $\theta = 60^\circ, 180^\circ, 300^\circ$

b. $\theta = 60^\circ, 300^\circ$

c. $\theta = 120^\circ, 240^\circ$

d. $\theta = 45^\circ, 135^\circ$

2 Simplify $\dfrac{1}{\sec\theta - \tan\theta}$.

a. $\sec\theta + \tan\theta$

b. $\sec\theta - \tan\theta$

c. $\cos\theta + \sin\theta$

d. 1

3 Given $\cot\theta = 2$ and θ is in Quadrant III, find the exact value of $\csc\theta$.

a. $-\sqrt{5}$

b. $\sqrt{5}$

c. $-\frac{\sqrt{5}}{2}$

d. $-\frac{1}{2}$

Section 2: Compound Angle Formulas

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4 Expand $\sin(A+B)$ using the addition formula.

a. $\sin A\cos B+\cos A\sin B$

b. $\sin A\cos B-\cos A\sin B$

c. $\cos A\cos B-\sin A\sin B$

d. $\sin A\sin B+\cos A\cos B$

5 Find the exact value of $\tan(75^\circ)$ using compound angle identities.

a. $2+\sqrt{3}$

b. $2-\sqrt{3}$

c. $\sqrt{3}+1$

d. $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

6 If $\tan A=2$ and $\tan B=3$, find the exact value of $\tan(A+B)$.

a. -1

b. 1

c. 5

d. -5

Section 3: Double Angle Formulas

7 Express $\cos(2\theta)$ strictly in terms of $\sin\theta$.

a. $1-2\sin^2\theta$

b. $2\sin^2\theta-1$

c. $1-\sin^2\theta$

d. $2\cos^2\theta-1$

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8 Solve the equation $\sin(2x) = \sin(x)$ for $0 \le x \le 2\pi$.

a. $x = 0, \pi/3, \pi, 5\pi/3, 2\pi$

b. $x = 0, \pi, 2\pi$

c. $x = \pi/3, 5\pi/3$

d. $x = 0, \pi/2, \pi, 3\pi/2, 2\pi$

9 Prove that $\dfrac{\sin(2\theta)}{1+\cos(2\theta)} \equiv \tan\theta$. What is the denominator after substitution?

a. $2\cos^2\theta$

b. $2\sin^2\theta$

c. $1-\cos^2\theta$

d. $2\cos\theta$

Section 4: The $R\cos(\theta \pm \alpha)$ / $R\sin(\theta \pm \alpha)$ Method

10 Express $3\cos\theta + 4\sin\theta$ in the form $R\cos(\theta - \alpha)$ where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

a. $5\cos(\theta - 53.1^\circ)$

b. $5\cos(\theta - 36.9^\circ)$

c. $7\cos(\theta - 53.1^\circ)$

d. $5\cos(\theta + 53.1^\circ)$

11 Find the maximum value of the function $f(\theta) = 5\cos\theta - 12\sin\theta + 7$ and the smallest positive θ at which it occurs.

a. Maximum is 20

b. Maximum is 13

c. Maximum is 26

d. Maximum is 19

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12 Solve $3\cos x + 4\sin x = 2.5$ for $0^\circ \leq x \leq 360^\circ$.

a. $x = 113.1^\circ$ and $x = 353.1^\circ$

b. $x = 60^\circ$ and $x = 300^\circ$

c. $x = 90^\circ$ and $x = 270^\circ$

d. $x = 45^\circ$ and $x = 225^\circ$

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